

Domain adaptation for *Catalan–Basque* machine translation via synthetic data and continued fine-tuning

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1 Motivation & research questions

Catalan–Basque (CA–EU) machine translation remains highly under-resourced, particularly in specialized domains. Existing multilingual models often struggle with this language pair, while high-quality in-domain parallel corpora are extremely limited.

Core question: Given the scarcity of domain-specific parallel corpora, can synthetic bilingual data generated from monolingual resources significantly improve CA–EU machine translation?

- Can pivot-based corpus construction and back-translation generate effective training data for low-resource specialized domains?
- Does domain-specific fine-tuning improve in-domain translation quality over the general model?
- Is continued fine-tuning from a general-domain checkpoint better than direct domain-specific SFT?

2 Two domains, two quality criteria

LITERARY Quality depends on fluency, creativity and stylistic naturalness.	CLINICAL Quality depends on terminological accuracy and clinical clarity.
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3 Datasets

SOURCE CORPORA	
GENERAL AINA CA–EU Parallel 100k pairs	LITERARY A CTILC CA–ES ~30k pairs ES pivot → EU synthetic
LITERARY B EhuHac ES–EU 100k pairs ES pivot → CA synthetic	CLINICAL E3-RC EU Clinical ~1,300 pairs · ca2eu only

TRAIN · VALIDATION · TEST SPLITS		
GENERAL eu2ca & ca2eu90 / 5 / 5 % split		
TRAIN 87.2k 43,661 ca2eu · 43,568 eu2ca	VALIDATION 4.8k 2,388 ca2eu · 2,458 eu2ca	TEST 4.8k 2,412 ca2eu · 2,435 eu2ca
LITERARY eu2ca & ca2euavg 110 tok		
TRAIN 102k 75,444 ca2eu · 26,900 eu2ca	VALIDATION 5.7k 4,217 ca2eu · 1,468 eu2ca	TEST 5.7k 4,261 ca2eu · 1,426 eu2ca
CLINICAL ca2eu onlyavg 1078 tok		
TRAIN 1,174	VALIDATION 65	TEST 66

4 Synthetic data construction

LITERARY A: CA (CTILC)



LITERARY B: EU-ES (EHUHAC)



CLINICAL EU-CA (E3C)



Round-trip technique: for model fine-tuning, we reverse the translation direction, using machine-translated texts as the source language and original texts (high quality) as the target language.

5 Training setup

Fine-tuning model	HiTZ/Latxa-Qwen3-VL-8B-Instruct
Translation model	HiTZ/Latxa-Llama-3.1-8B-Instruct
Adaptation	LoRA · 4-bit NF4 (bitsandbytes)
Max length	768 tokens
Epochs	3
Seed	42
Batch	4
Grad accum	8
Learning rate	5e-5 · cosine schedule
Checkpoint selection	Best BLEU on validation

Literary domain: each parallel pair → two instruction samples eu2ca + ca2eu .
Clinical domain: single direction (due to data limitations) ca2eu .

6 Model variants & experimental design

GENERAL – DIRECT SFT ON GENERAL-DOMAIN PARALLEL DATA



LITERARYV1 – DIRECT LITERARY SFT (SYNTHETIC PARALLEL DATA THROUGH PIVOTING)



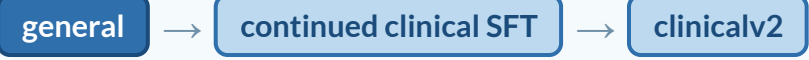
LITERARYV2 – CONTINUED FROM GENERAL



CLINICALV1 – DIRECT CLINICAL SFT



CLINICALV2 – CONTINUED FROM GENERAL



7 General domain results (bidirectional)

MODEL	DIR	CHRF++↑	BLEU↑	TER↓	COMET↑
Baseline	eu→ca	10.93	0.60	300.3	45.17
Baseline	ca→eu	17.10	1.29	296.3	48.54
generalv1	eu→ca	45.03	18.51	75.53	80.61
generalv1	ca→eu	41.63	9.92	88.02	80.75

Baseline: Latxa-Qwen3-VL-8B-Instruct zero-shot.

8 Literary domain results (bidirectional)

MODEL	DIR	CHRF++↑	BLEU↑	TER↓	COMET↑
Baseline	eu→ca	11.96	0.82	151.3	41.55
Baseline	ca→eu	16.59	0.63	266.6	44.82
generalv1	eu→ca	27.97	4.96	89.44	66.68
generalv1	ca→eu	29.58	2.82	101.8	67.76
literaryv1 ★	eu→ca	36.13	8.96	85.08	69.61
literaryv1 ★	ca→eu	26.90	2.44	99.60	65.29
literaryv2	eu→ca	34.51	7.44	87.66	68.72
literaryv2	ca→eu	25.87	2.31	100.7	64.34

★ Direct SFT (v1) beats continued adaptation (v2).

9 Clinical domain results (ca→eu)

MODEL	CHRF++↑	BLEU↑	TER↓	COMET↑
Baseline	11.03	0.64	104.3	50.56
generalv1	10.60	0.38	98.10	56.56
clinicalv1 ★	40.20	19.43	101.09	76.25
clinicalv2	38.73	18.50	104.64	75.02

Clinicalv2 achieves best chrF++ & BLEU and lowest TER; clinicalv1 leads on COMET.

10 Token-matched results (ca→eu)

MODEL	CHRF++↑	BLEU↑	TER↓	COMET↑
clinicalv1 ★	40.20	19.43	101.09	76.25
literaryv1	24.73	2.28	97.03	65.41

Each model is trained on the same amount of data (~360,000 tokens) and evaluated on their respective domains.

11 Key findings

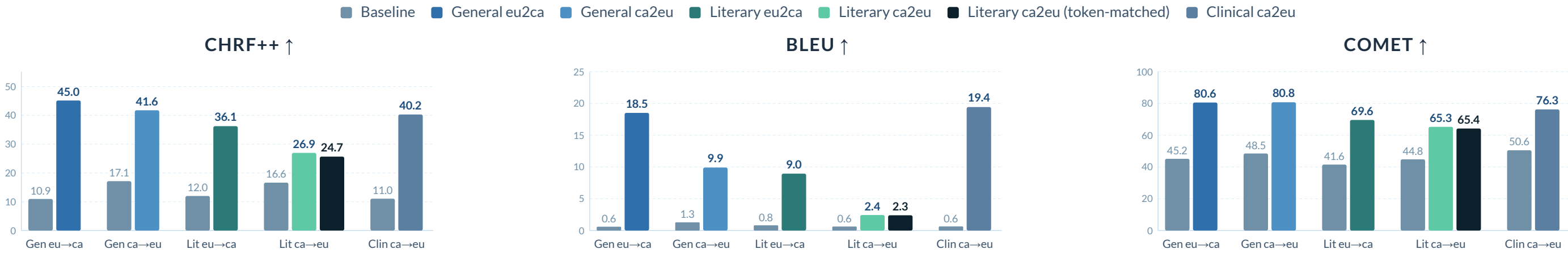
- Domain-specific fine-tuning is the main driver of improvement, in particular for clinicalv1.
- General-domain transfer through continued SFT does not enhance domain-specific performance: direct SFT remains stronger.
- Generating training pairs either via a pivot language (ES) or direct translation successfully compensates for the absolute lack of parallel corpora.
- Clinical domains are more "learnable": under strict token-matching, clinical SFT yields for most metrics substantially higher results than literary SFT.

12 Limitations & future work

Limitations: Small clinical test set (N=66); single-directional clinical training; reliance on synthetic supervision.

CORPUS Due to high reliance on synthetic data, explore SFT results on human-translated parallel data. Extend clinical to eu2ca.	ABLATIONS Test all possible pivot language choices and SFT data sizes variation between domains. Try English prompting.
EVALUATION Incorporate human evaluation.	SINGLE MODEL Single-architecture setup limits generalization across model families.

Best model per domain vs. zero-shot



RESOURCES

projecte-aina/CA-EU_Parallel_Corpus · EhuHac EU literary corpus · CTILC CA literary corpus · E3C clinical EU corpus

BASE MODEL (FINE-TUNING)

HiTZ/Latxa-Qwen3-VL-8B-Instruct

TRANSLATION MODEL (DATA CONSTRUCTION)

HiTZ/Latxa-Llama-3.1-8B-Instruct

GITHUB REPOSITORY

